Quiz-3

ACOL 216: Introduction to Computer Architecture

IIT Delhi - Abu Dhabi · Semester II, 2025-26

(Answer all the questions with proper explanations.)

Total Marks: 20 **Time: 60 minutes**

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1. Consider a non-pipelined processor with a clock rate of 2.5 GHz and an average cycles per instruction (CPI) of 4. The same processor is upgraded to a pipelined architecture with 5 stages. However, due to internal pipeline delays, the clock rate is reduced to 2 GHz. Assume that there are no pipeline stalls.
   (i) Calculate the total execution time for 150 instructions in the non-pipelined processor.
   (ii) Calculate the total execution time for 150 instructions in the pipelined processor.
   (iii) Determine the speedup achieved by the pipelined processor over the non-pipelined processor.

   (3 + 3 + 2 marks)

2. Consider a pipelined processor with the following stages:
   - IF: Instruction Fetch
   - ID: Instruction Decode and Operand Fetch
   - EX: Execute
   - WB: Write Back

   The IF, ID, and WB stages take one clock cycle each. The number of clock cycles required for the EX stage depends on the instruction type:
   - ADD and SUB instructions require 2 clock cycle in the EX stage.
   - MUL instruction requires 4 clock cycles in the EX stage.

   Assume that operand forwarding is used in the pipeline.

   Determine the number of clock cycles required to complete the execution of the following instruction sequence:

```
ADD R2, R1, R0    // R2 <- R0 + R1
MUL R4, R3, R2    // R4 <- R3 * R2
SUB R6, R5, R4    // R6 <- R5 - R4
```

   (3 marks)

3. Consider a 4-stage pipelined processor with stages S_1, S_2, S_3, and S_4. The number of clock cycles required by four instructions I_1, I_2, I_3, and I_4 in each stage is given in the table below:

	S_1	S_2	S_3	S_4
I_1	2	1	1	1
I_2	1	3	2	2
I_3	2	1	1	3
I_4	1	2	2	2

   Determine the total number of clock cycles required to complete the execution of the instruction sequence $I_1, I_2, I_3,, I_4$.

   (3 marks)

4. The baseline execution time of a program on a 2 GHz single-core machine is 100 nanoseconds (ns). The portion of the code corresponding to 90% of the execution time can be fully parallelized. The overhead for using each additional core is 10 ns when running on a multicore system. Assume that all cores execute their share of the parallelized portion equally. Determine the number of cores that minimizes the total execution time of the program. **(3 marks)**

5. Consider a 3-stage pipelined processor with stage delays of 10 ns, 20 ns, and 14 ns for the first, second, and third stages, respectively. Assume that there are no additional delays and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle. Determine the total execution time (in nanoseconds) required to execute 100 instructions on this processor. **(3 marks)**